JAVA SNACKS – SERUBAWON
(Thursday, July 9, 2026)

Instructions:

1. Avoid using Artificial Intelligence(AI) tools.

2. Submission must occur before class (8:31am – Friday, July 10, 2026).

3. Each question is to be in a different file.

Submission Guideline: Create the folder serubawon and save all of the files inside it

Compress folder into a zip file and send into my DM.

1. Write a Java program that declares three integer variables firstInteger, secondInteger, and thirdInteger, assigns them the values 10, 20, and 30, and prints each value on a separate line.
2. Write a program that reads a student's name and age from the user and prints: 'Hello, [name]. You are [age] years old.'
3. Write a sequence of statements that converts a temperature from Celsius to Fahrenheit using the formula $F = (C \times 9/5) + 32$ and prints the result.
4. Explain in your own words what a sequence statement is and give one real-life example of a sequence of instructions.
5. Trace through the following sequence and write down what is printed:

```
int number = 5;
number  = number + 3;
number  = number  * 2;
System.out.println(number );
```

6. Write a program that reads two numbers from the user, computes their sum, difference, product, and quotient, and prints each result on a separate labelled line.
7. A program contains these statements in order:

   (1) Read price,

   (2) Calculate tax = price × 0.075,

   (3) Calculate total = price + tax,

   (4) Print total.

   Write these four statements in Java.
8. Write a sequence of statements that reads a radius from the user, computes the area of a circle (area = ℗ × r^2), and prints the result to 2 decimal places using printf.
9. Trace through the following sequence and state the final value of each variable:

```
int firstNumber = 4;
int secondNumber= 6;
int thirdNumber = firstNumber  + secondNumber;
firstNumber = thirdNumber  – 2;
 secondNumber = firstNumber  * 3;
```

Website: www.semicolon.africa Tel: 0906 000 8609

Email: info@semicolon.africa Social: @semicolonafrica

10. Write a program that reads a person's first name, last name, and year of birth. Compute their age (assume current year is 2025) and print a formatted profile.
11. Identify the error in this sequence and explain why it will not compile:

    System.out.println(total);

    double total = price + tax;
12. Write a sequence that declares a String variable called greeting, assigns it the value 'Good morning!', prints it, then reassigns it to 'Good evening!' and prints it again.
13. A cashier's till must: (1) Read item price, (2) Read quantity, (3) Calculate subtotal = price × quantity, (4) Calculate VAT = subtotal × 0.20, (5) Calculate grand total = subtotal + VAT, (6) Print grand total. Write these six statements in Java.
14. What is the difference between declaring a variable and initialising a variable? Give a Java example of each.
15. Write a program that reads a distance in miles, converts it to kilometres (1 mile = 1.60934 km), and prints both values clearly labelled.
16. Write a sequence of statements that swaps the values of two integer variables firstNumber and secondNumber using a temporary variable. Print the values before and after the swap.
17. Trace the following sequence and state what is printed: String name = "Java"; int length = name.length(); System.out.println(name + " has " + length + " letters.");
18. Write a program that reads a student's score out of 50, scales it to a mark out of 100, and prints both the original and scaled marks.
19. Explain what would happen if you tried to print a variable before it has been assigned a value. What type of error would this cause in Java?
20. Write a sequence of five statements that:

    (1) declares a double called balance set to 5000.00,

    (2) deposits 1200.50,

    (3) withdraws 750.25,

    (4) applies 1.5% monthly interest,

    (5) prints the final balance to 2 decimal places.
21. Write an if statement that reads an integer from the user and prints: 'Positive' if it is greater than zero.
22. Write an if/else statement that reads an integer and prints 'Even' if it is divisible by 2, or 'Odd' otherwise.
23. Write an if/else if/else chain that reads a score (0–100) and prints the letter grade: A (90–100), B (80–89), C (70–79), D (60–69), F (below 60).
24. Explain the difference between an if/else statement and an if/else if/else chain. When would you use one over the other?
25. Trace through the following selection and write down what is printed when x = 15: if (x > 20) System.out.println("Big"); else if (x > 10) System.out.println("Medium"); else System.out.println("Small");
26. Write a selection statement that reads a person's age. Print 'Child' if age < 13, 'Teenager' if 13–17, 'Adult' if 18–64, and 'Senior' if 65 or above.
27. Write a program that reads two integers and uses if/else to print: the larger of the two, the smaller, and whether they are equal.
28. Explain why the following selection produces a logic error and correct it: int x = 10; if (x = 10) System.out.println("Ten");
29. Write a nested if statement that reads a username and password. If the username is 'admin' AND the password is '1234', print 'Access granted', otherwise print 'Access denied'.

30. Write a selection that reads an integer between 1 and 7 and prints the corresponding day of the week (1 = Monday, 7 = Sunday). Handle invalid input.
31. A utility company charges: ₦50/unit for 0–100 units, ₦75/unit for 101–300 units, and ₦100/unit for above 300 units. Write a selection statement that reads the number of units and prints the bill.
32. Trace through the following selection when a = 5 and b = 10, and state what is printed: if (a > b) System.out.println("A wins"); else if (a == b) System.out.println("Draw"); else System.out.println("B wins");
33. Write a selection statement that reads three integers and prints the largest of the three without using Math.max().
34. Explain the difference between == and .equals() when comparing values in Java. Give an example showing where using == would give an incorrect result.
35. Write an if/else statement that reads a year and prints whether it is a leap year. A year is a leap year if divisible by 4 but not 100, unless also divisible by 400.
36. Write a selection that reads a character and prints: 'Vowel' if it is a, e, i, o, or u (upper or lower case), 'Consonant' if it is any other letter, or 'Not a letter' otherwise.
37. What is short-circuit evaluation in Java? Give an example using && and explain how it prevents a runtime error.
38. Write a selection statement that reads two doubles and an operation code (+, -, *, /). Perform the correct operation and print the result. Handle division by zero.
39. Trace the following and state the output when score = 72: String grade; if (score >= 90) grade = "A"; else if (score >= 80) grade = "B"; else if (score >= 70) grade = "C"; else grade = "F"; System.out.println("Grade: " + grade);
40. Write a selection statement that reads a person's income and computes their tax: 0% on the first ₦300,000; 7% on ₦300,001–₦600,000; 15% on anything above ₦600,000. Print the total tax owed.
41. What is a dangling else problem? Illustrate with a code example and explain how to resolve it using braces.
42. Write a selection that reads an integer and prints: 'Divisible by 3 and 5' if divisible by both, 'Divisible by 3 only', 'Divisible by 5 only', or 'Divisible by neither'.
43. Write a program that reads three sides of a triangle and uses selection statements to determine whether it is Equilateral, Isosceles, Scalene, or Invalid (does not form a valid triangle).
44. Explain what happens if you use a single if statement three times instead of an if/else if/else chain when only one condition can be true at a time. Why is if/else if/else preferred in such cases?
45. Write a selection statement that reads a month number (1–12) and prints the number of days in that month. For February, ask the user for the year and account for leap years.
46. Write a for loop that prints the integers from 1 to 10, each on its own line.
47. Write a while loop that prints the integers from 10 down to 1.
48. Write a do-while loop that prompts the user to enter a positive number. Keep prompting until a positive value is entered, then print it.
49. Explain the difference between a while loop and a do-while loop. Give an example situation where a do-while loop is more appropriate.
50. Write a for loop that computes and prints the sum of integers from 1 to 100.
51. Trace through the following loop and write down exactly what is printed: for (int count = 1; count <= 5; count++) { System.out.println(count * count); }
52. Write a while loop that reads integers from the user until the user enters 0, and prints the total sum of all entered numbers.
53. Explain what an infinite loop is. Write an example of one and explain how to fix it.

54. Write a for loop that prints all even numbers between 1 and 50 on one line, separated by spaces.
55. Write a for loop that prints the multiplication table of a number entered by the user (from 1× to 12×).
56. Trace the following loop and state the final value of count after it finishes: int count = 0; int counter = 1; while (counter <= 20) { if (counter % 3 == 0) count++; counter++; }
57. Write a do-while loop that presents the user with a menu (1=Add, 2=Subtract, 3=Exit) and processes their choice. The menu keeps appearing until the user enters 3.
58. Write a for loop that computes n! (n factorial) for a value of n entered by the user. Print the result.
59. Explain the difference between break and continue inside a loop. Write one example for each.
60. Write a while loop that reads a series of exam scores (ending when the user enters -1) and prints the highest score, lowest score, and average.
61. Write a for loop that prints the following pattern for n = 5: * ** *** **** *****
62. What is a sentinel value? Write a while loop that uses the sentinel value -999 to end a series of number inputs and prints the count of numbers entered.
63. Write a for loop that reads 10 integers and counts how many are positive, how many are negative, and how many are zero.
64. Trace the following loop when number = 3 and write down every line printed:

```
for (int count = 1; count <= number; count ++) {
    for (int subCount = 1; subCount  <=count ; subCount ++) {
        System.out.print(subCount  + " ");
    }
            System.out.println();
}
```

65. Write a while loop that finds and prints all factors of a positive integer entered by the user.
66. Explain what a loop variable (counter) is and what happens if you forget to update it inside a while loop.
67. Write a for loop that computes the sum of the series: 1/1 + 1/2 + 1/3 + ... + 1/n, where n is entered by the user. Print the result to 4 decimal places.
68. Write a do-while loop that simulates a simple number guessing game: generate a random number between 1 and 10, prompt the user to guess, and keep looping until they guess correctly. Count the attempts.
69. Explain the three parts of a for loop header (initialisation, condition, update). What happens if the condition is always true? What if it is false from the start?
70. Write a for loop that prints all prime numbers between 2 and 50. For each candidate, use a nested loop or condition to check divisibility.
71. Write a program that uses a loop to collect 10 exam scores from the user. Use selection inside the loop to classify each score as Pass (≥50) or Fail (<50). After the loop, print the total number of passes and fails.
72. Trace through the following combined sequence, selection, and iteration and write down every line printed:

```
for (int count  = 1; count  <= 5; count ++) {
    if (count  % 2 == 0) System.out.println(count + " is even");
    else System.out.println(count  + " is odd");
}
```

73. Write a program that uses a while loop to keep reading positive integers from the user (stopping when 0 is entered). Inside the loop, use selection to print 'Prime' or 'Not prime' for each number.
74. Write a program using a loop and selection to print the FizzBuzz sequence from 1 to 50: print 'Fizz' if divisible by 3, 'Buzz' if divisible by 5, 'FizzBuzz' if divisible by both, or the number otherwise.
75. Explain how sequence, selection, and iteration each contribute to a program that reads 5 student marks and prints the grade for each. Write such a program.
76. Write a do-while menu-driven program with options: 1=Deposit, 2=Withdraw, 3=Check Balance, 4=Exit. Use selection inside the loop to handle each option. Prevent withdrawals that exceed the current balance.
77. Write a program that uses a for loop to display a number pattern. Inside the loop, use selection to print each number in the range 1–20 as 'Low' (1–7), 'Mid' (8–14), or 'High' (15–20) next to the number.
78. Trace the following code and write down every line of output:

```
int total = 0;
for (int count  = 1; count  <= 4; count ++) {
        if (count  % 2 != 0) total += count ;
}
System.out.println("Total: " + total);
```

79. Write a program that uses a while loop to simulate a bank ATM: keep presenting a menu until the user exits. Use selection to deposit, withdraw (with overdraft protection), or display the balance.
80. Write a program that reads an integer n and uses nested for loops with selection to print: all pairs (i, j) where $1 \leq i < j \leq n$ and i + j is even.
81. Explain the difference between a pre-test loop (while/for) and a post-test loop (do-while). Give a scenario where each is the better choice and justify your answer.
82. Write a program that uses a for loop to read 10 integers. Inside the loop, use selection to count how many are: divisible by 2 only, divisible by 3 only, divisible by both, divisible by neither. Print the four counts after the loop.
83. Write a program that uses a do-while loop to keep prompting the user for a password until they enter the correct one ('secret123'). After 3 failed attempts, print 'Account locked.' and stop.
84. Trace the following nested structure and write what is printed:

```
for (int count = 3; count >= 1; count--) {
        for (int counter = 1; counter <= count; counter++) {
                System.out.print("* ");
            }
            System.out.println();
    }
```

85. Write a program that reads a positive integer n and uses a loop with selection to determine whether n is prime, printing all divisors found and whether the number is prime or not.
86. Write a program that uses iteration and selection to print a salary slip: read an employee's basic salary, use selection to calculate tax (0% ≤ ₦50k, 10% ≤ ₦150k, 20% above), subtract tax, and print gross, tax, and net pay. Repeat for 5 employees using a loop.
87. Identify and explain three different errors (one syntax, one logic, one runtime) that could each occur in a program combining loops and selection statements. Give a code example of each.
88. Write a program using a for loop that reads n pairs of numbers and for each pair uses selection to print the larger one. After the loop, print which pair had the overall largest number.

89. Write a program that uses a while loop to simulate rolling a die (random number 1–6) until a 6 is rolled. Count and print the number of rolls it took. After the game ends, use selection to print a comment: fewer than 5 rolls → 'Lucky!', 5–10 → 'Average', more than 10 → 'Tough luck!'
90. Explain with a code example how the same problem (printing numbers 1–10) can be solved using a for loop, a while loop, and a do-while loop. Discuss which form is most readable for this problem and why.
91. A program must read student scores repeatedly until the user enters -1, then print the count, sum, average, highest, and lowest scores. Write the complete program using appropriate sequence, selection, and iteration. Justify your choice of loop type.
92. Explain the concept of loop invariant. Choose a simple while loop (e.g., computing a sum) and identify what stays true before and after each iteration.
93. Write a program that uses nested loops and selection to validate a 9×9 Sudoku row entered by the user as 9 integers: check that every number is between 1 and 9 and that no number repeats. Print 'Valid row' or list all errors found.
94. Trace through the following complex combination and write every line of output. Then rewrite it using a for loop instead of the while loop: int n = 1, sum = 0; while (n <= 10) { if (n % 2 == 0) { sum += n; System.out.println("Added: " + n); } n++; } System.out.println("Sum: " + sum);
95. Write a complete program that simulates a number guessing game: generate a random number between 1 and 100, allow the user up to 10 guesses, give 'Too high' or 'Too low' feedback, and after the game print their score (10 points minus one point per wrong guess). Use sequence, selection, and iteration appropriately.
96. Explain the difference between a counted loop and a conditional loop. Give a programming scenario for each and explain which type of loop is better suited and why.
97. Write a program that reads a positive integer and uses iteration and selection to print its prime factorisation. For example, 360 → 2 × 2 × 2 × 3 × 3 × 5. Explain how your loop condition and selection statement work together.
98. Discuss the concept of loop control flow. Explain with code examples how break, continue, and a flag variable each provide different ways to exit or skip iterations. Identify a scenario where each is the most appropriate choice.
99. Write a menu-driven program that runs until the user chooses Exit. The menu offers: (1) Check if a number is prime, (2) Find all factors of a number, (3) Print a multiplication table, (4) Exit. Each option uses appropriate iteration and selection. All inputs must be validated.
100. Explain, using pseudocode and a corresponding Java program, how sequence, selection, and iteration combine to solve the following problem: Read 10 employee records (name and hourly wage), compute each employee's weekly pay (hours × wage, with time-and-a-half for hours above 40), classify each as 'Low' (<₦50,000), 'Mid' (₦50,000–₦150,000), or 'High' (>₦150,000), and print a formatted payroll report at the end.

Have fun guys!

Rest and Recover!

Do all our tasks!